CHROMA: Colour Intelligence—Bridging Physical Colours and Digital Design

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***Abstract*— Transferring colours from physical surfaces into digital design applications is complicated by illumination, surface reflectance, sensor response and fragmented software workflows. This paper presents CHROMA, a prototype integrating CHROMAPOD, a portable ESP32–TCS34725 colour scanner, with CHROMA Studio, a mobile-compatible colour analysis application. CHROMAPOD supports acquisition, local CIE76 comparison and Bluetooth Low Energy (BLE) exchange; the companion application provides format conversion, CIEDE2000 image matching, surface recolouring and colour-reference management. Calibration was evaluated on 16 printed RGB reference patches, with five complete scans per patch for each of two firmware versions (80 scans per version). Mean channel-wise RGB agreement against intended digital codes increased from 85.02% in V13 to 94.68% in V14, a 9.66 percentage-point improvement. These values quantify agreement with nominal digital references under the experimental conditions, not certified colourimetric accuracy, because the printed swatches were not independently characterized and some reference colours informed calibration development. The results demonstrate a physical–digital workflow requiring independent validation.**

***Index Terms*— colour sensing, ESP32, TCS34725, BLE, CIELAB, CIEDE2000, colour visualization, calibration**

I. Introduction

A colour seen on a physical surface must first be measured before it can be reused reliably in digital design. Visual judgement and phone photographs vary with lighting and image processing [10]–[12]. CHROMA connects a portable ESP32/TCS34725 scanner (CHROMAPOD) to a companion application (CHROMA Studio) for colour comparison, conversion, image matching, visualization and saved references. Scanned colours travel to the app over BLE, while image-match results return to the device. This paper describes the implemented workflow and evaluates a firmware calibration refinement using printed references, with the limits of that evaluation made explicit.

II. Related Work and Existing Systems

Colour comparison first requires a numerical description of colour. CIELAB represents lightness (L^*) and two chromatic dimensions (a^* and b^*); CIE76 measures their Euclidean separation, while CIEDE2000 adjusts for perceptual differences [1], [2]. Research on colour transfer and colorization informs photographic editing [3], [4], although CHROMA does not claim to implement those exact methods. The TCS34725 returns red, green, blue and clear responses [5]; the ESP32 supplies I^2C, SPI and BLE [15]. An RGBC sensor cannot recover full spectral reflectance; CIELAB is standardized in ISO/CIE 11664-4:2019 [6]. Nix Mini 3 and Datacolor ColorReader provide mobile-linked physical sensing [7], [8], while Pantone Connect emphasizes digital palettes and conversions [9]. CHROMA is assessed as an educational integration, not as a demonstrated accuracy improvement over those products.

III. System Architecture and Hardware

CHROMA follows the path shown in Fig. 1: a surface is scanned by CHROMAPOD, processed and displayed locally, and optionally sent over BLE to CHROMA Studio for digital analysis. This separates physical acquisition from image processing while keeping the workflow connected. The ESP32-WROOM-32 communicates with the TCS34725 via I^2C, drives a 1.8-inch ST7735 TFT using SPI, and monitors a KY-040 rotary encoder and dedicated scan button. The illumination LED on the sensor module and a shroud help reduce ambient-light variation. A rechargeable 3.7-V lithium-polymer battery, charging/protection module and implemented regulated 5-V stage supply the development board; the board distributes 3.3

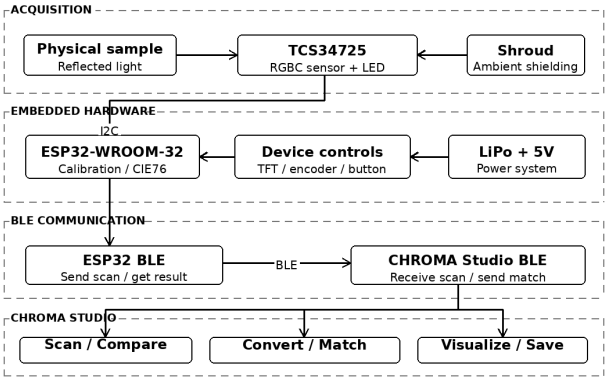


Fig. 1. CHROMA architecture from physical RGBC acquisition and embedded processing to bidirectional BLE and companion-app functions.

V to peripherals. Fig. 2 details the actual circuit, and Table I lists the corresponding pins; Fig. 3 shows the assembled device. Together, the illustrations distinguish the system workflow from its physical implementation.

Table I summarizes the I^2C sensor, SPI display and input connections illustrated in Fig. 2. The scan button starts acquisition, the encoder selects modes, and the TFT shows local results. BLE connects image matching and reference storage. A shared ground and regulated supply serve the integrated prototype.

TABLE I
SELECTED CHROMAPOD SIGNAL CONNECTIONS

Component	Signal	ESP32 connection
TCS34725	SDA / SCL	GPIO21 / GPIO22
ST7735	SCK / MOSI	GPIO18 / GPIO23
ST7735	CS / DC / RST	GPIO27 / GPIO26 / GPIO25
KY-040	CLK / DT / SW	GPIO33 / GPIO14 / GPIO13
Scan button	Input	GPIO32 (pull-up)
Power	Boost 5 V / ground	VIN / GND

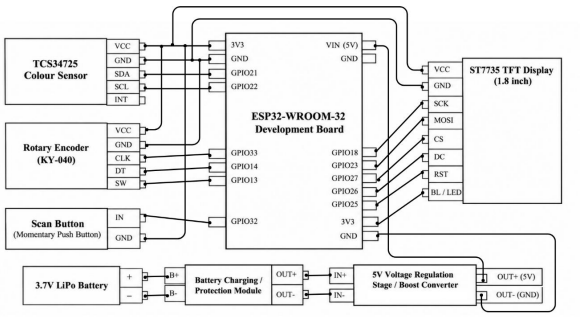


Fig. 2. CHROMAPOD component and power connections, corresponding to the implemented ESP32 pin assignments.

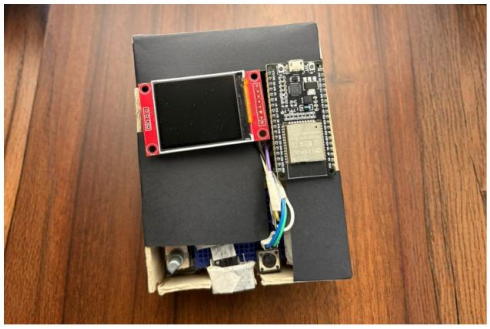


Fig. 3. Assembled CHROMAPOD prototype showing its TFT and ESP32 hardware.

The scanner offers SCAN, COMPARE and IMAGE MATCH modes. SCAN presents measured colours and numerical values locally; COMPARE compares two acquired references by CIE76; IMAGE MATCH passes the scanned colour to the application for comparison with a selected image reference and displays the returned CIEDE2000 result. CHROMA Studio also provides Convert, Visualize Colour, Library and Projects functions. The software is a responsive web application packaged for Android using Capacitor; BLE exchange is handled at the device/application boundary.

IV. Colour Acquisition and Processing

A. Acquisition and Normalization

A single sensor reading may fluctuate even when the surface remains unchanged. CHROMAPOD therefore averages five successive RGBC samples within each completed scan. For channel x, this mean is

$$\bar{x} = \frac{1}{5}\sum_{i=1}^{5} x_i. \tag{1}$$

The averaged channels still depend on the sensor's black and white response. Normalization places each channel on a common zero-to-one scale, with clipping to keep outputs within that range:

$$n_x = \mathrm{clip}\left(\frac{x - B_x}{W_x - B_x}, 0, 1\right), \tag{2}$$

where B_x and W_x are the calibrated reference levels for the same channel. V13 RGBC black references were $(71.10, 49.80, 28.70, 150.10)$ and white references $(715.067, 557.133, 328.600, 1617.867)$; these are setup-specific calibration values. Degenerate $W_x - B_x$ would invalidate normalization.

B. Neutral and Chromatic Corrections

Grey-like samples need different handling from strongly coloured surfaces to avoid unwanted colour casts. The firmware measures the spread of normalized RGB channels as

$$d_{\mathrm{RGB}} = \max(n_r, n_g, n_b) - \min(n_r, n_g, n_b). \tag{3}$$

A sample is considered approximately neutral when $d_{\mathrm{RGB}} < 0.04$. For near-neutral samples, the normalized clear-channel value is used with a fitted tone exponent, $t = n_c^{0.544}$, and special handling near black. Chromatic samples instead undergo a calibrated 3×3 correction:

$$\begin{bmatrix} r' \\ g' \\ b' \end{bmatrix} = \mathbf{M} \begin{bmatrix} n_r \\ n_g \\ n_b \end{bmatrix}, \tag{4}$$

where $\mathbf{M}$ denotes an empirically fitted matrix. The resulting linear channels are clipped and encoded using the IEC-defined sRGB transfer function [13], $f(c) = 12.92c$ for $c \le 0.0031308$ and $f(c) = 1.055c^{1/2.4} - 0.055$ otherwise. Channels are then scaled to 8-bit RGB. The model targets the specified reference setup and should not be interpreted as spectral reconstruction or cross-device colour certification.

C. Colour Difference and Image Matching

For two physical scans, COMPARE reports the numerical separation of their CIELAB values. This is calculated on the device using CIE76:

$$\Delta E_{76} = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}. \tag{5}$$

The user interface maps this numerical difference to an indicative match percentage using $\mathrm{clip}(\mathrm{round}(100-2\Delta E_{76}),0,100)$. This is a display heuristic, not a standard perceptual probability.

For a scanned colour and an image-selected colour, the companion app instead uses CIEDE2000, which accounts more closely for perceptual variation [1], [2]:

$$\Delta E_{00} = \left[\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{k_C S_C}\right)\left(\frac{\Delta H'}{k_H S_H}\right) \right]^{1/2}. \quad (6)$$

The application computes the result and returns the selected colour and difference to the ESP32 over BLE. CIE76 and CIEDE2000 values are derived by different models and should not be numerically interchanged.

V. Companion Application

CHROMA Studio extends a physical scan into a digital design workflow. After receiving a colour, users can inspect it through Scan or Compare, view RGB/HEX/HSL/CMYK/CIELAB representations through Convert, and reuse it through Library and Projects (Fig. 4). Image Match compares it with a colour selected from a photograph and returns the result over BLE (Fig. 5). To preview a change on a photographed surface, Visualize Colour lets the user select a region automatically or manually and refine it using Brush and Eraser tools. The selected or saved colour is applied to that region while attempting to retain luminance variation and surface texture. Before, After and Slider comparisons and an adjustable intensity control support design exploration. Material categories include wall/paint, fabric, wood, plastic, metal and ceramic. The recoloured image is a photographic preview, not a radiometrically validated prediction of how a manufactured surface will appear. Region-aware editing has a substantial research history, including gradient-domain techniques [14]; CHROMA does not claim to implement that specific algorithm. This separation enables a scanned reference to be reused across different design tasks without repeating the physical measurement.

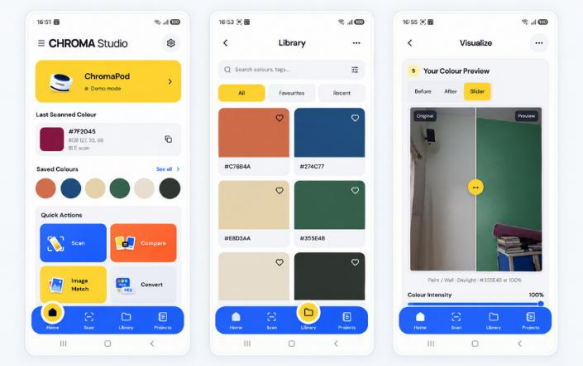
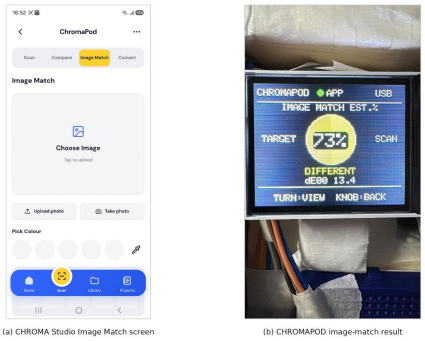


Fig. 4. CHROMA Studio screens for scanning, library management and visual colour preview.



(a) CHROMA Studio Image Match screen (b) CHROMAPOD image-match result

Fig. 5. Image Match application interface and the corresponding result shown on the hardware display.

In Fig. 5, CHROMAPOD sends the scanned colour to the application, which returns the selected image colour and CIEDE2000 difference over BLE. The device handles acquisition and local display, whereas CHROMA Studio handles image processing. Figure 6 presents the before/after recolouring preview, retaining visible shading without claiming physically accurate material appearance.

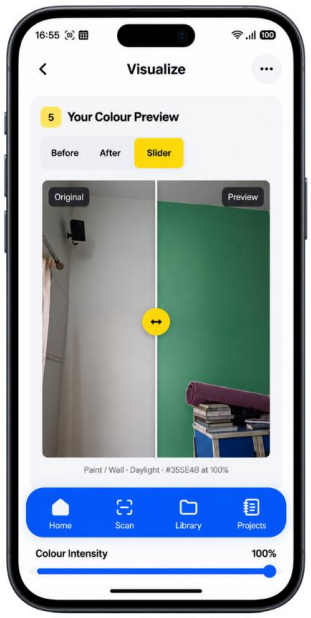


Fig. 6. Before/after visual preview of a recoloured image region in CHROMA Studio.

VI. Experimental Method

A. Printed Reference Procedure

To evaluate the improvement in colour agreement from V13 to V14, both firmware versions were tested using an identical printed reference sheet. As shown in Fig. 7, the sheet contains sixteen patches: four neutral references (white, light grey, mid-grey and black) and twelve chromatic references (red, orange, yellow, lime, green, teal, cyan, blue, indigo, purple, pink and brown). The TCS34725 was positioned over each patch using the prototype's illuminated and shrouded scanning arrangement.

Five independent completed scans were recorded per patch for V13 and repeated for V14, giving $16 \times 5 = 80$ scan records per version and 160 overall. Each completed scan averages five sensor samples (Eq. 1); the repeated scans give an initial assessment of colour-wise response and short-term variation.

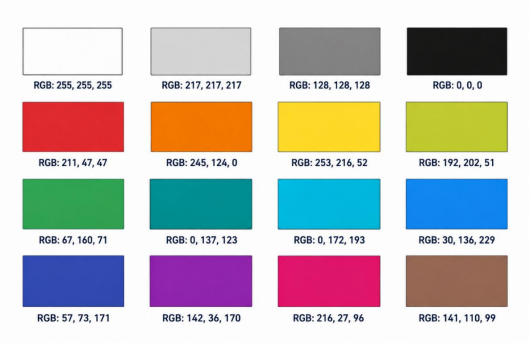


Fig. 7. Sixteen printed reference patches used for calibration evaluation.

The numbers printed beside the patches are intended digital RGB values, not independently measured paper colours. To compare the scanner outputs with those nominal targets, the study used channel-wise agreement. For measured channels (R_m, G_m, B_m) and intended reference (R_t, G_t, B_t), agreement was computed as

$$A = 100\left(1 - \frac{|R_m - R_t| + |G_m - G_t| + |B_m - B_t|}{765}\right)\%. \tag{7}$$

The overall score averages agreement over scan records. This is a direct channel-wise metric, not ΔE_{76} or ΔE_{00}. Moreover, a consumer print does not necessarily reproduce its nominal RGB input; precise print-to-screen reproduction requires colour-managed profiles and viewing assumptions [16]. No spectrophotometric measurements of the printed patches were available. Some references also informed calibration fitting; consequently this is an internal calibration comparison, not independent generalization testing.

B. Functional Checks

Functional evaluation included mode navigation, scanning and display updates, CIE76 comparison, BLE exchange during IMAGE MATCH, image colour selection, visualization on a photographed wall region and saved-reference workflows. The report documents these as selected functionality demonstrations rather than an exhaustive automated validation. No independent measurement of battery endurance, absolute colour difference against certified samples or multi-device transfer accuracy is asserted here.

VII. Results and Discussion

Several firmware versions were developed before V13; V13 was selected as the documented evaluation baseline, not the first working release. Its colour deviations motivated targeted refinements in V14. The V13 baseline yielded 85.02% mean RGB agreement; after V14 calibration refinement, mean agreement increased to 94.68%, an absolute gain of 9.66 percentage points (Table II). The changes involved black/white normalization, chromatic correction and near-neutral handling. Table II summarizes the within-prototype improvement. It measures closeness to nominal RGB codes under the tested conditions, not the true spectral colour of the printed paper.

TABLE II
Calibration comparison on printed references

Measure	V13	V14
Printed reference colours	16	16
Complete scans per colour	5	5
Total complete scans	80	80
Mean RGB agreement	85.02%	94.68%
V14 improvement: +9.66 percentage points		

For representative neutrals, V13 mid-grey outputs were $(145, 145, 145)$, $(144, 144, 144)$ twice, and $(143, 143, 143)$ twice against $(128, 128, 128)$. V14 measurements were $(127, 127, 127)$, $(125, 125, 125)$, $(124, 124, 124)$ twice, and $(122, 122, 122)$, showing reduced channel discrepancy. All V14 black records were $(0, 0, 0)$ and all white records were $(255, 255, 255)$, but endpoint clipping prevents these identical outputs from establishing identical raw sensor responses. Light-grey outputs ranged from 208 to 213 versus a target of 217.

The overall mean hides variation between hues. Figure 8 shows V14 agreement by colour: lime reached 98.61%, indigo 87.56%, red 94.14%, blue 90.43%, purple 95.35% and pink 95.14%. Figure 9 places these results alongside the V13 profile. The graphs show that stronger average agreement does not imply equally strong performance for every hue. Repeated scans showed limited variation for some patches (five 8-bit levels for mid grey and light grey), but more visible variation for some chromatic channels (e.g., pink green-channel outputs from 0 to 11).

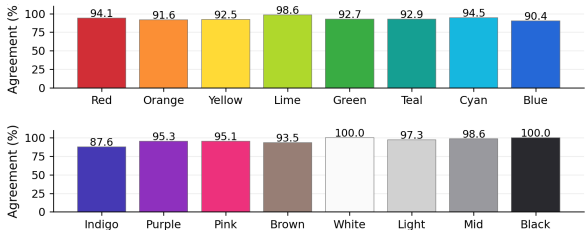


Fig. 8. V14 per-colour RGB agreement across the sixteen printed references.

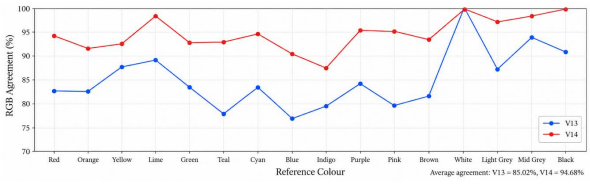


Fig. 9. Comparison of V13 and V14 colour-wise agreement profiles; plotted figures reproduce the project's measured evaluation results.

The scans suggest short-term consistency, not independent accuracy or long-term repeatability.

A. Neutral and Chromatic Response

Mid-grey readings moved from 143–145 in V13 to 122–127 in V14 against a target of 128. Light grey still measured 208–213 against 217; calibration improved agreement without eliminating tone errors. Identical encoded black and white outputs do not prove identical raw readings because values may be clipped. The 11.05-point gap between lime (98.61%) and indigo (87.56%) confirms that performance differs by hue; the reference sheet cannot isolate lighting, surface and sensor contributions.

B. Interpretation of Colour Agreement

Equation 7 averages absolute encoded RGB-channel errors; it is reproducible but not perceptually uniform. Equal channel errors can appear different at different brightness levels. Unlike the CIE76 and CIEDE2000 measures used for application comparisons, the experiment's percentage describes agreement with nominal printed-sheet codes, not instrument-measured colour difference.

C. Threats to Validity

Nominal printed RGB codes were not checked with a spectrophotometer, and some swatches informed calibration fitting. Lighting, print medium and contact positioning may also favour the chosen setup. These results therefore document a within-prototype improvement, not performance on unseen materials. Reproduction should record sensor gain, integration time, lighting, print medium, positioning and firmware version.

VIII. Limitations and Future Work

The TCS34725 measures broad RGBC responses rather than full reflectance spectra. Image recolouring is a design preview, not a certified material-colour prediction. Cross-device variation, long-term drift, BLE reliability and battery endurance were not independently evaluated.

A. Independent Validation

Future studies should calibrate against characterized samples under controlled lighting, freeze calibration parameters, then test independent colours and materials. Saving raw RGBC and encoded RGB would reveal output clipping; instrument-referenced errors and repeated sessions would establish more meaningful performance limits [11], [12]. Painted panels, textiles and glossy surfaces should be included.

B. Design and Communication Implications

Per-colour results should accompany global agreement so hue-dependent errors remain visible. Future improvements include guided calibration, a compact PCB and enclosure, and design-tool integration. BLE tests should separately assess reconnection, message-result association and reference persistence rather than treating software operation as evidence of optical accuracy.

IX. Conclusion

CHROMA demonstrates a working integration of ESP32/TCS34725 colour acquisition and a BLE-linked digital application for numerical comparison, conversion, image matching, visualization and reference management. A five-scan-per-colour comparison on 16 printed references found that mean RGB agreement with nominal codes increased from 85.02% (V13) to 94.68% (V14). The 9.66 percentage-point gain supports the effectiveness of the calibration refinement on this evaluation set, while its non-certified and partially calibration-linked references constrain interpretation. The contribution is the demonstrated physical-to-digital workflow and a documented, appropriately limited calibration improvement.

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